$e^- \leftarrow A \rightarrow B$

$E = \frac{1}{2} m_e v_e^2 + m_e c^2 \qquad v_e = \sqrt{\frac{2(E - m_e c^2)}{m_e}}$

$m_e v_e = m_B v_B \qquad v_B = \frac{m_e}{m_B} v_e$

$$m_A c^2 = \underline{m_e c^2} + m_B c^2 + \frac{1}{2} m_B v_B^2 + \underline{\frac{1}{2} m_e v_e^2}$$

$$E = m_A c^2 - m_B c^2 - \frac{1}{2} m_B v_B^2 = m_A c^2 - m_B c^2 - \frac{1}{2} \frac{m_e^2}{m_B} v_e^2$$

$$= m_A c^2 - m_B c^2 - \frac{m_e}{m_B}(E - m_e c^2) = m_A c^2 - m_B c^2 - \frac{m_e}{m_B} E + \frac{m_e^2}{m_B} c^2$$

$$\therefore \left(1 + \frac{m_e}{m_B}\right) E = c^2 \left(m_A - m_B + \frac{m_e^2}{m_B}\right)$$

$$(m_B + m_e) E = c^2 (m_A m_B - m_B^2 + m_e^2)$$

$$E = \frac{m_A m_B - m_B^2 + m_e^2}{m_B + m_e} c^2$$